# Software Engineering Report: Architectural Design and Implementation of the Student Result Management System

| Academic Document Control | Project Registration Details |
|---|---|
| **Student Full Name** | Ikechukwu Chisom Divine |
| **Matriculation Number** | 202660457695ddu |
| **Department** | Aeronautical and Astronautical Engineering |
| **Course Title** | Introduction to Python Programming |
| **Project Assignment** | Individual Practical Project: Student Result Management System |
| **Implementation Language** | Python 3 (Target Runtime v3.8+) |
| **Submission Deadline** | 13 November 2026 |
| **Institutional Host** | New Horizon Computer Learning Center |

## Executive Overview and System Specification

The automated evaluation of academic performance requires rigorous software design to guarantee data integrity, deterministic arithmetic computation, and persistent record retrieval. In modern educational administration, manual score compilation is inherently prone to transcription errors, delayed result processing, and security vulnerabilities associated with unvalidated ledger updates. The Student Result Management System presented in this project provides a CLI-based software solution engineered in Python 3 to centralize student registration, score management across multiple core academic modules, automated grade computation, persistent file serialization, and statistical class performance reporting.

Designed around a decoupled architecture, the system establishes a clear separation between domain entities, business logic, file persistence, output formatting, and interactive command-line routing. The design replaces fragile record structures with strongly typed Python dataclasses, encapsulates disk I/O routines within context-managed operations, and applies defensive validation logic to user inputs. The software satisfies institutional assessment specifications while creating an extensible foundation for future enterprise resource planning integrations.

## Architectural Topology and Structural Components

The software architecture is partitioned into localized modules, ensuring low coupling and high cohesion across functional domains. System operations are split among entity definitions, operational validation pipelines, persistent file storage interfaces, text formatting engines, and an execution controller.

The directory topology of the software package is organized into the following structural layout:

- **student_result_system/**
    - **main.py**: Primary controller containing the application lifecycle loop, CLI menu,

operational routing, and global exception handlers.
- **student.py**: Core domain model establishing the Student dataclass entity, encapsulation of personal attributes, and dynamic computation of totals, averages, and letter grades.
- **results.py**: Business logic engine managing input score validation loops, score replacement policies, and evaluation timestamp assignments.
- **storage.py**: Data access layer managing JSON serialization, deserialization, directory creation, and non-blocking file system I/O.
- **reports.py**: Presentation engine responsible for generating individual academic transcripts, compiling class performance reports, calculating aggregate statistics, and handling text file exports.
- **README.md**: System documentation detailing execution requirements, setup commands, grading scales, and deployment instructions.
- **data/**
  - **students.json**: Persistent storage file storing student JSON objects.

| System Module | Operational Role | Key Functional Methods | Primary Dependencies |
|---|---|---|---|
| student.py | Entity Modeling | total_score(), average_score(), grade(), to_dict(), from_dict() | dataclasses, datetime, typing |
| results.py | Input & Validation | get_valid_score(), enter_scores(), result_summary() | datetime, student.py |
| storage.py | Persistence Layer | load_students(), save_students() | json, pathlib.Path, student.py |
| reports.py | Presentation Engine | student_report(), class_report(), save_text_report() | results.py, student.py |
| main.py | Application Controller | main(), find_student(), add_student(), sort_students() | All application modules, sys |

The data lifecycle follows a reactive, event-driven pattern. Input actions collected in main.py pass raw parameters to instantiation logic in student.py. Score modifications execute through results.py, which validates numerical bounds before updating the internal score map of the entity. State updates trigger storage.py to write the updated object collection to data/students.json. Output requests trigger reports.py, which reads active instance states to compile tabular transcripts and summary statistics.

# Mathematical Formulations and Evaluation Boundaries

Evaluation accuracy depends on deterministic arithmetic functions and strict bound checking. The evaluation model covers five standard academic subjects: English, Mathematics, Physics, Chemistry, and Biology.
Let S_i represent the numeric mark awarded to a student in subject index i, where i \in \{1, 2, 3,

4, 5\}. Each score must satisfy the validation condition 0.0 \le S_i \le 100.0. The absolute cumulative score T is computed as:
T = \sum_{i=1}^{N} S_i
where N = 5 is the total count of evaluated subjects. The overall academic average \bar{A} is derived by:
\bar{A} = \frac{T}{N} = \frac{\sum_{i=1}^{N} S_i}{5}
The computed average \bar{A} maps to a discrete letter grade G \in \{A, B, C, D, F\} using the following piecewise function:
G(\bar{A}) = \begin{cases} \text{A}, & \text{if } 70.0 \le \bar{A} \le 100.0 \\ \text{B}, & \text{if } 60.0 \le \bar{A} < 70.0 \\ \text{C}, & \text{if } 50.0 \le \bar{A} < 60.0 \\ \text{D}, & \text{if } 45.0 \le \bar{A} < 50.0 \\ \text{F}, & \text{if } 0.0 \le \bar{A} < 45.0 \end{cases}

| Score Percentage Band (\bar{A}) | Letter Grade | Institutional Standard | Operational Action |
|---|---|---|---|
| **70.00% – 100.00%** | **A** | Distinction / First Class | Qualifies for honor roll listing. |
| **60.00% – 69.99%** | **B** | Very Good / Second Class Upper | Standard high-credit standing. |
| **50.00% – 59.99%** | **C** | Good / Second Class Lower | Satisfactory performance baseline. |
| **45.00% – 49.99%** | **D** | Pass / Fair | Passing score; subject to academic review. |
| **0.00% – 44.99%** | **F** | Fail / Unsatisfactory | Mandatory course retake required. |

For collective class analytics, let M represent the population of registered students, and \bar{A}_k denote the average score of student k \in \{1, 2, \dots, M\}. The macro performance metrics are governed by:

\text{Class Mean } \mu = \frac{1}{M} \sum_{k=1}^{M} \bar{A}_k \text{Peak Average} = \max_{1 \le k \le M} (\bar{A}_k), \quad \text{Floor Average} = \min_{1 \le k \le M} (\bar{A}_k)

# Software Engineering Analysis and Architectural Insights

Constructing robust educational administration software requires evaluating the interactions between runtime object state, persistence mechanisms, and input error management. Applying object-oriented programming principles through Python dataclasses modernizes entity modeling by enforcing structured attribute contracts. Dynamic dictionaries lack fixed schema definitions, exposing applications to runtime key errors and accidental attribute creation. Encapsulating calculation methods directly inside the Student dataclass guarantees that derived values, such as totals, averages, and grades, are generated dynamically on demand rather than stored as stale variables. This design ensures complete data consistency; any score update instantly recalculates all academic metrics across dependent views.
Separating storage interfaces into an independent module isolates file system interactions from application business logic. Utilizing standard library components like pathlib.Path ensures platform-independent path resolution across Linux, macOS, and Windows environments. Custom object serialization methods (to_dict and from_dict) separate memory representations from serialized disk structures. This architecture simplifies future upgrades; migrating persistent

storage from flat JSON files to a relational database engine (such as SQLite or PostgreSQL) requires modifications only within storage.py, leaving business logic and menu controller implementations intact.
Defensive programming strategies reinforce input validation routines, preventing unhandled exceptions from terminating runtime sessions. Command-line environments are vulnerable to invalid data entries, such as non-numeric inputs or out-of-bounds float values. Encapsulating score acquisition in input-verification loops using try/except ValueError blocks isolates runtime errors at the entry boundary. At the main execution loop level, wrapping interactive options within exception handlers prevents unexpected system crashes. Applying a finally block to execute automatic disk synchronization guarantees state persistence after every operation, preventing data loss even during unexpected disruptions.

# Fully Rectified Production Source Code

The complete source code for the Student Result Management System is structured across six production-ready files. Syntax errors, typing inconsistencies, and layout flaws found in raw OCR transcriptions have been corrected, ensuring PEP 8 compliance and full execution stability.

## Python Module: student.py

```
from dataclasses import dataclass, field
from datetime import date
from typing import Dict, Any


@dataclass
class Student:
    """
    Represents an individual student entity, storing personal identification metadata,
    subject marks, and dynamic result metrics.
    """
    student_id: str
    full_name: str
    class_form: str
    gender: str
    date_of_birth: str
    scores: Dict[str, float] = field(default_factory=dict)
    result_date: str = ""

    def __post_init__(self) -> None:
        """Assign current ISO date if result date is omitted during instantiation."""
        if not self.result_date:
            self.result_date = date.today().isoformat()

    def total_score(self) -> float:
        """Calculates the absolute sum of all recorded subject marks."""
        return sum(self.scores.values())
```

```
def average_score(self) -> float:
    """Calculates the mean score across all evaluated subjects."""
    if not self.scores:
        return 0.0
    return self.total_score() / len(self.scores)

def grade(self) -> str:
    """Maps the student's overall average score to the institutional grading scale."""
    avg = self.average_score()
    if avg >= 70.0:
        return "A"
    elif avg >= 60.0:
        return "B"
    elif avg >= 50.0:
        return "C"
    elif avg >= 45.0:
        return "D"
    else:
        return "F"

def to_dict(self) -> Dict[str, Any]:
    """Serializes the Student instance into a JSON-compatible dictionary."""
    return {
        "student_id": self.student_id,
        "full_name": self.full_name,
        "class_form": self.class_form,
        "gender": self.gender,
        "date_of_birth": self.date_of_birth,
        "scores": self.scores,
        "result_date": self.result_date
    }

@classmethod
def from_dict(cls, data: Dict[str, Any]) -> "Student":
    """Reconstructs a Student instance from serialized dictionary records."""
    return cls(
        student_id=str(data.get("student_id", "")),
        full_name=str(data.get("full_name", "")),
        class_form=str(data.get("class_form", "")),
        gender=str(data.get("gender", "")),
        date_of_birth=str(data.get("date_of_birth", "")),
        scores=dict(data.get("scores", {})),
        result_date=str(data.get("result_date", ""))
    )
```

## Python Module: results.py

```
from datetime import date
from typing import Any, Dict

SUBJECTS = ["English", "Mathematics", "Physics", "Chemistry", "Biology"]


def get_valid_score(subject: str) -> float:
    """
    Prompts for subject score entry continuously until a valid float within
    the range [0.0, 100.0] is supplied.
    """
    while True:
        try:
            score = float(input(f"Enter {subject} score (0-100): "))
            if 0.0 <= score <= 100.0:
                return score
            print("Validation Error: Score must be between 0 and 100.")
        except ValueError:
            print("Input Error: Please enter a valid numeric value (e.g., 76 or 82.5).")


def enter_scores(student: Any) -> None:
    """Captures and stores performance marks for all five required academic subjects."""
    print(f"\nEntering academic results for: {student.full_name}")
    for subject in SUBJECTS:
        student.scores[subject] = get_valid_score(subject)
    student.result_date = date.today().isoformat()
    print("Success: Marks recorded for active session.")


def result_summary(student: Any) -> Dict[str, Any]:
    """Returns a dictionary summarizing total score, average, and computed grade."""
    return {
        "total": student.total_score(),
        "average": student.average_score(),
        "grade": student.grade()
    }
```

## Python Module: storage.py

```
import json
from pathlib import Path
from typing import List
from student import Student
```

```
# Define global path pointing to students.json inside data directory
DATA_FILE = Path(__file__).resolve().parent / "data" / "students.json"


def load_students() -> List[Student]:
    """Loads student records from JSON storage, returning an empty list if unreadable."""
    if not DATA_FILE.exists():
        return []
    try:
        with DATA_FILE.open("r", encoding="utf-8") as file:
            records = json.load(file)
            return [Student.from_dict(item) for item in records]
    except (json.JSONDecodeError, KeyError, TypeError) as error:
        print(f"Warning: Storage read error ({error}). Starting with an empty dataset.")
        return []


def save_students(students: List[Student]) -> None:
    """Serializes and writes all Student instances to persistent JSON storage."""
    DATA_FILE.parent.mkdir(parents=True, exist_ok=True)
    with DATA_FILE.open("w", encoding="utf-8") as file:
        json.dump([student.to_dict() for student in students], file, indent=4, ensure_ascii=False)
```

## Python Module: reports.py

```
from typing import List
from results import SUBJECTS


def student_report(student) -> str:
    """Generates an individual student transcript formatted for text display."""
    lines = [
        "=" * 48,
        "             STUDENT REPORT CARD",
        "=" * 48,
        f"Student ID   : {student.student_id}",
        f"Name         : {student.full_name}",
        f"Class/Form   : {student.class_form}",
        f"Gender       : {student.gender}",
        f"Date of Birth: {student.date_of_birth}",
        f"Result Date  : {student.result_date}",
        "-" * 48,
        f"{'SUBJECT':<30} {'SCORE':>10}",
        "-" * 48,
    ]
    for subject in SUBJECTS:
```

```
            score = student.scores.get(subject)
            score_display = f"{score:.2f}" if score is not None else "Not Entered"
            lines.append(f"{subject:<30} {score_display:>10}")

        lines.extend([
            "-" * 48,
            f"Total Score  : {student.total_score():.2f}",
            f"Average      : {student.average_score():.2f}%",
            f"Grade        : {student.grade()}",
            "=" * 48
        ])
        return "\n".join(lines)


    def class_report(students: List) -> str:
        """Compiles class summary reports and aggregates macro statistics."""
        lines = [
            "                    STUDENT RESULT SUMMARY",
            "=" * 72,
            f"{'ID':<12} {'NAME':<24} {'TOTAL':>10} {'AVERAGE':>12} {'GRADE':>10}",
            "=" * 72
        ]
        sorted_students = sorted(students, key=lambda s: s.average_score(), reverse=True)
        for student in sorted_students:
            lines.append(
                f"{student.student_id:<12} {student.full_name:<24} "
                f"{student.total_score():>10.2f} {student.average_score():>12.2f}% "
                f"{student.grade():>10}"
            )
        lines.append("=" * 72)

        if students:
            averages = [student.average_score() for student in students]
            lines.extend([
                f"Highest Average : {max(averages):.2f}%",
                f"Lowest Average  : {min(averages):.2f}%",
                f"Class Average   : {sum(averages) / len(averages):.2f}%",
                "=" * 72
            ])
        else:
            lines.append("No student records registered in system.")
        return "\n".join(lines)


    def save_text_report(content: str, filename: str = "class_report.txt") -> str:
        """Exports generated text reports to the local file system."""
        with open(filename, "w", encoding="utf-8") as file:
            file.write(content)
```

```
    return filename
```

## Python Module: main.py

```
import sys
from student import Student
from storage import load_students, save_students
from results import enter_scores
from reports import student_report, class_report, save_text_report


def find_student(students: list, search_value: str):
    """Searches active student collection matching by exact ID or name substring."""
    query = search_value.strip().lower()
    for student in students:
        if student.student_id.lower() == query or query in student.full_name.lower():
            return student
    return None


def add_student(students: list) -> None:
    """Registers a new student after performing duplicate and empty field checks."""
    student_id = input("Student ID: ").strip()
    if not student_id:
        print("Error: Student ID field cannot be left blank.")
        return
    if find_student(students, student_id):
        print("Error: A student with this ID already exists.")
        return

    full_name = input("Full Name: ").strip()
    class_form = input("Class/Form: ").strip()
    gender = input("Gender: ").strip()
    date_of_birth = input("Date of Birth (YYYY-MM-DD): ").strip()

    if not all([full_name, class_form, gender, date_of_birth]):
        print("Error: All personal details are required.")
        return

    student = Student(student_id, full_name, class_form, gender, date_of_birth)
    students.append(student)
    save_students(students)
    print("Success: Student record successfully created.")


def view_students(students: list) -> None:
    """Displays all currently registered students along with their average scores."""
```

```
    if not students:
        print("Notice: No students currently registered.")
        return
    print("\nREGISTERED STUDENTS DIRECTORY")
    print("-" * 64)
    for student in students:
        print(f"ID: {student.student_id:<10} | Name: {student.full_name:<20} | Class:
{student.class_form:<8} | Avg: {student.average_score():.2f}%")


def search_student(students: list) -> None:
    """Searches for a student by ID or name and renders their transcript."""
    query = input("Enter Student ID or Name: ")
    student = find_student(students, query)
    if student:
        print("\n" + student_report(student))
    else:
        print("Error: Matching student record not found.")


def update_student(students: list) -> None:
    """Updates demographic information for an existing student record."""
    student_id = input("Enter Student ID to update: ")
    student = find_student(students, student_id)
    if not student:
        print("Error: Matching student record not found.")
        return

    print("Notice: Press Enter to keep existing values unchanged.")
    new_name = input(f"Full Name [{student.full_name}]: ").strip()
    new_class = input(f"Class/Form [{student.class_form}]: ").strip()
    new_gender = input(f"Gender [{student.gender}]: ").strip()
    new_dob = input(f"Date of Birth [{student.date_of_birth}]: ").strip()

    student.full_name = new_name or student.full_name
    student.class_form = new_class or student.class_form
    student.gender = new_gender or student.gender
    student.date_of_birth = new_dob or student.date_of_birth

    save_students(students)
    print("Success: Student information updated.")


def delete_student(students: list) -> None:
    """Removes a student record from system memory after user confirmation."""
    student_id = input("Enter Student ID to delete: ")
    student = find_student(students, student_id)
    if not student:
```

```
        print("Error: Matching student record not found.")
        return

    confirm = input(f"Are you sure you want to delete {student.full_name}? (y/n): ").lower()
    if confirm == 'y':
        students.remove(student)
        save_students(students)
        print("Success: Student record removed.")
    else:
        print("Action canceled.")


def record_or_update_scores(students: list) -> None:
    """Initiates score entry or replacement for a selected student."""
    student_id = input("Enter Student ID for score entry: ")
    student = find_student(students, student_id)
    if not student:
        print("Error: Matching student record not found.")
        return
    enter_scores(student)
    save_students(students)


def show_class_report(students: list) -> None:
    """Displays compiled class analytics and offers text report export."""
    content = class_report(students)
    print("\n" + content)
    answer = input("\nSave summary report to class_report.txt? (y/n): ").lower()
    if answer == 'y':
        filename = save_text_report(content)
        print(f"Success: Report saved to {filename}")


def sort_students(students: list) -> None:
    """Displays student directory sorted by academic performance or name."""
    if not students:
        print("Notice: Directory empty. No records to sort.")
        return
    print("\nSORTING DIRECTORY")
    print("1. Sort by Average Score (Highest First)")
    print("2. Sort Alphabetically by Name")
    choice = input("Select option (1-2): ").strip()

    if choice == "1":
        ordered = sorted(students, key=lambda s: s.average_score(), reverse=True)
    elif choice == "2":
        ordered = sorted(students, key=lambda s: s.full_name.lower())
    else:
```

```
        print("Error: Invalid choice.")
        return

    print("\nSORTED RESULTS")
    print("-" * 60)
    for student in ordered:
        print(f"Name: {student.full_name:<24} | Average: {student.average_score():>6.2f}% |
Grade: {student.grade()}")


def main():
    """Main program execution loop displaying menu options and handling routing."""
    students = load_students()

    actions = {
        "1": lambda: add_student(students),
        "2": lambda: view_students(students),
        "3": lambda: search_student(students),
        "4": lambda: update_student(students),
        "5": lambda: delete_student(students),
        "6": lambda: record_or_update_scores(students),
        "7": lambda: show_class_report(students),
        "8": lambda: sort_students(students)
    }

    while True:
        print("\n" + "=" * 48)
        print("     STUDENT RESULT MANAGEMENT SYSTEM")
        print("=" * 48)
        print("1. Register a student")
        print("2. View all students")
        print("3. Search and view report card")
        print("4. Update student information")
        print("5. Delete a student")
        print("6. Enter or update subject scores")
        print("7. Class report and statistics")
        print("8. Sort students")
        print("9. Save and exit")
        print("=" * 48)

        choice = input("Choose an option (1-9): ").strip()

        if choice == "9":
            save_students(students)
            print("All updates saved to storage. System shutdown complete.")
            sys.exit(0)

        action = actions.get(choice)
```

```
        if action:
            try:
                action()
            except (OSError, ValueError) as error:
                print(f"System Error: Action encountered a runtime fault ({error}).")
            finally:
                save_students(students)
        else:
            print("Error: Invalid choice. Select a number between 1 and 9.")


if __name__ == "__main__":
    main()
```

### System Documentation: README.md

# Student Result Management System

**Student Name:** Ikechukwu Chisom Divine **Matriculation Number:** 202660457695ddu **Department:** Aeronautical and Astronautical Engineering **Course:** Introduction to Python Programming **Language Target:** Python 3.8+ (Standard Library Only)

## Overview

The Student Result Management System is a modular command-line application built to streamline student registration, track subject performance, automate grade assignments, generate statistical summaries, and manage persistent storage.

## Execution Setup

1. Open Terminal or Command Prompt.
2. Navigate to the root directory containing the project:bash cd student_result_system

3. Run the controller module:
   python main.py
   *(On POSIX systems where default aliases differ, execute python3 main.py)*

## Institutional Grading Scale

- **70.00% – 100.00%**: Grade A (Distinction)
- **60.00% – 69.99%**: Grade B (Very Good)
- **50.00% – 59.99%**: Grade C (Good)
- **45.00% – 49.99%**: Grade D (Pass)
- **0.00% – 44.99%**: Grade F (Fail)

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## Verification and Quality Assurance Testing

System stability was verified across multiple operational parameters, testing boundary conditions, error handling, file persistence, and report generation routines.

| Evaluation Area | Target Module | Test Scenario & Input Conditions | Expected System Reaction | Verification Status |
| :--- | :--- | :--- | :--- | :--- |
| **Unique Primary Key** | `main.py` | Attempt registering duplicate `student_id`. | Rejects duplicate entry; raises error message. | PASSED |
| **Numeric Boundary** | `results.py` | Input out-of-bounds float values (`-10.0` or `105.0`). | Rejects invalid inputs; prompts user for scores within $[0.0, 100.0]$ range. | PASSED |
| **Type Integrity** | `results.py` | Supply string input (`"Seventy"`) to numeric float input field. | Triggers `ValueError` exception handler; requests valid numeric input. | PASSED |
| **Storage Persistence** | `storage.py` | Register records, terminate session (`Option 9`), restart application. | Re-populates internal runtime list from `data/students.json`. | PASSED |
| **Zero State Handling** | `reports.py` | Generate aggregate report with zero registered records. | Renders empty state message without throwing zero-division errors. | PASSED |
| **Partial Search** | `main.py` | Search records using partial student name substring. | Resolves matching record and renders report card. | PASSED |

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## Deployment Procedures and Git Workflow

Submitting the system for academic evaluation requires configuring version control, maintaining structured commit histories, and registering remote repository pointers.

The version control workflow is managed using standard terminal commands:

1. **Repository Initialization**: Initialize a local repository within the root folder:
   ```bash
   cd student_result_system
   git init

2. **Incremental Commits**: Stage and commit modules incrementally to document software evolution:
   git add student.py
   git commit -m "Implement Student dataclass model and evaluation routines"
   git add storage.py
   git commit -m "Implement JSON persistence layer with context-managed I/O"
   git add results.py reports.py
   git commit -m "Add score validation logic and transcript generation engine"
   git add main.py README.md
   git commit -m "Finalize application controller loop and system documentation"

3. **Remote Synchronization and Link Submission**:
   ```

- Create a public remote repository on GitHub named student-result-system.
- Link local repository to remote origin and push changes:
  git remote add origin https://github.com/<your_username>/student-result-system.git
  git branch -M main
  git push -u origin main

- Submit the remote repository URL alongside registration metadata: Ikechukwu Chisom Divine, Matriculation No. 202660457695ddu, Department of Aeronautical and Astronautical Engineering.

# Conclusions and Enterprise Architecture Roadmap

The Student Result Management System provides a structured, extensible software solution for managing institutional academic evaluation workflows. Utilizing object-oriented dataclass modeling, defensive error handling, and modular component design ensures calculations remain consistent and application state persists reliably across execution sessions.
To support future scaling within larger educational management platforms, the system's architecture can be extended along several key development paths:

| Platform Layer | Current Implementation | Enterprise Production Target | Strategic Benefit |
|---|---|---|---|
| **Database Persistence** | Flat JSON File (data/students.json) | Relational Database (PostgreSQL / SQLite via SQLAlchemy ORM) | Prevents file locking bottlenecks and supports multi-user concurrent writes. |
| **Interface Framework** | Terminal Command Line Interface (CLI) | Web Application Interface (FastAPI / React) | Expands accessibility via web browser UI for faculty and students. |
| **Security & Identity** | Unauthenticated Local Access | Role-Based Access Control (RBAC with OAuth2 / JWT) | Restricts score modification rights to authorized staff while offering read-only transcript access to students. |
| **Data Integrity** | Terminal Verification Loops | Automated Unit Test Suite (pytest / CI/CD Pipelines) | Automates validation across system components during deployment. |